

Comparison between ARCH and LSTM for Hong Kong Volatility Forecasting

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Github: https://github.com/JamesManuel/ECON4130_Term_Paper.git

Abstract

This paper compares the performance of a traditional ARCH volatility model and a Long Short-Term Memory (LSTM) neural network in forecasting equity market volatility in Hong Kong. Using daily Hang Seng Index data from May 2002 to November 2025, we construct 21-day realized log volatility as the forecast target and evaluate model performance out of sample over the period 2022–2025. An ARCH(7) specification is employed to flexibly capture short-term volatility persistence, while the LSTM provides a nonlinear autoregressive setup using the same information set. Forecast accuracy is assessed using MAE, MSE, and QLIKE loss functions.

The results show that both models deliver very similar forecasting performance across all metrics. Although the LSTM achieves marginally lower losses, the differences relative to the ARCH model are small. Overall, the findings indicate that for monthly-horizon volatility forecasting at the index level, a carefully specified linear ARCH model performs comparably to a more complex neural network. This suggests that increased model complexity does not necessarily yield substantial gains in predictive accuracy in this setting.

1. Introduction

Daily stock returns are typically close to serially uncorrelated with a conditional mean near zero, whereas volatility displays strong persistence and is therefore forecastable (*Engle, 1982; Bollerslev, 1986*). Accurate volatility forecasts play a central role in risk management, derivatives pricing, margin setting, and volatility-based trading strategies (*Andersen & Bollerslev, 1998*). Despite the global importance of the Hong Kong equity market, relatively little empirical work has examined volatility forecasting for this market using modern machine learning methods, particularly in direct comparison with well-specified traditional econometric models.

This study compares two approaches that rely on the same historical information from the Hang Seng Index (HSI): an ARCH(7) model representing a flexible linear volatility framework and a Long Short-Term Memory (LSTM) neural network that allows for nonlinear autoregressive dynamics. Both models are tasked with forecasting 21-day realized log volatility of the HSI, with performance evaluated out of sample using MAE, MSE, and QLIKE loss functions. The objective is not to demonstrate the superiority of machine learning methods, but to assess whether the additional complexity of a neural network yields material improvements in forecast accuracy relative to a conventional ARCH specification in the context of Hong Kong index volatility.

2. Data

2.1 Dataset

We rely on daily closing prices of the Hang Seng Index to compute log returns and realized volatility. The dataset spans from 14 May 2002 through 8 November 2025, providing roughly twenty-three and a half years of daily observations. To preserve a realistic forecasting setting, the data are split

chronologically so that the models are trained only on information that would have been available at the time of prediction. The training sample runs from 2002 to the end of 2021, totalling 4,847 observations, while the out-of-sample period begins on 1 January 2022 and continues through November 2025, contributing 944 observations. In total, the full dataset contains 5,791 daily data points.

Set	Date	Observations
Full Sample Set	2002-05-14 to 2025-11-08	5791
Training Set	2002-05-14 to 2021-12-31	4847
Out-of-Sample Set	2022-01-01 to 2025-11-08	944

The decision to evaluate forecasts specifically over 2022–2025 serves several purposes. First, it enforces a clean forward-looking setup that avoids any look-ahead bias by withholding all post-2021 information during model estimation, consistent with how volatility forecasting is implemented in practice. Second, this window includes a series of markedly different market conditions in Hong Kong, ranging from post-pandemic normalization to shifting macroeconomic pressures tied to China and Hong Kong, as well as abrupt volatility spikes and subsequent reversals. Finally, a multi-year horizon provides a more meaningful test of model robustness by examining predictive stability across both calm and turbulent market regimes rather than relying on a brief or idiosyncratic volatility episode.

3. Methodology

3.1 Forecast Target: 21-Day Realized Log Volatility

We define daily log-returns as:

$$r_t = \ln(P_t) - \ln(P_{t-1})$$

Realized variance (RV) over the next 21 trading days:

$$RV_{t,t+21} = \sum_{i=1}^{21} r_{t+i}^2$$

We forecast the logarithm of realized variance:

$$y_t = \frac{1}{2} \log(RV_{t,t+21})$$

We construct 21-day realized log volatility as the forecast target, following the realized volatility literature (*Andersen et al., 2003; Barndorff-Nielsen & Shephard, 2002*). This forecast horizon aligns with typical option maturities and reduces measurement noise relative to daily volatility.

To ensure strict out-of-sample evaluation, all models use only data available up to time t when predicting y_t .

3.2 ARCH(7) Model

To benchmark the LSTM against a traditional econometric volatility model while avoiding excessive persistence imposed by a GARCH structure, we employ an ARCH(7) specification (*Engle, 1982*). This choice allows conditional variance to depend flexibly on the past seven daily squared returns, aligning the model more closely with the 21-day realized volatility target while avoiding overly smooth variance dynamics.

The return process is specified as:

$$r_t = \sigma_t \varepsilon_t, \varepsilon_t \sim \mathcal{N}(0,1)$$

with conditional variance:

$$\sigma_t^2 = \omega + \sum_{i=1}^7 \alpha_i r_{t-i}^2$$

Parameters are estimated using data up to 31 December 2021. Multi-step-ahead variance forecasts are constructed recursively and aggregated to form a 21-day ahead realized variance forecast, which is then log-transformed to match the target variable. All forecasts are produced strictly out of sample from January 2022 onward, ensuring a realistic forecasting environment.

The 21-day ahead realized variance forecast is given by:

$$\widehat{RV}_{t,t+21}^{\text{ARCH}} = \sum_{i=1}^{21} \hat{\sigma}_{t+i|t}^2,$$

where $\hat{\sigma}_{t+i|t}^2$ denotes the ARCH-implied conditional variance for day $t+i$ based on information available at time t . Returns are modeled in percent, so the conditional variance forecasts correspond to squared percent returns. To express the variance in terms of original log returns, the forecast is rescaled by 100^2 :

$$\widehat{RV}_{t,t+21}^{\text{ARCH, unscaled}} = \frac{1}{100^2} \sum_{i=1}^{21} \hat{\sigma}_{t+i|t}^2.$$

Finally, the forecast of 21-day realized log volatility is constructed as:

$$\hat{y}_t^{\text{ARCH}} = \frac{1}{2} \log \left(\widehat{RV}_{t,t+21}^{\text{ARCH, unscaled}} \right) = \frac{1}{2} \log \left(\frac{1}{100^2} \sum_{i=1}^{21} \hat{\sigma}_{t+i|t}^2 \right).$$

3.3 LSTM Neural Network

Input Design and Supervised Learning Setup

We use a Long Short-Term Memory (LSTM) neural network (*Hochreiter & Schmidhuber, 1997*) to model nonlinear autoregressive volatility dynamics. LSTM-based models have been widely applied to financial time series forecasting (*Fischer & Krauss, 2018*). The LSTM model is designed to forecast 21-day realized log volatility using a rich set of volatility-related features over multiple horizons constructed from daily Hang Seng Index returns rather than relying solely on lagged realized volatility.

To account for temporal dependence, the supervised learning problem is formulated using rolling sequences of length 120 trading days. For each forecast origin t , the input consists of a sequence of past feature vectors, and the model produces a one-step-ahead forecast of future 21-day log volatility:

$$\hat{y}_t^{\text{LSTM}} = f(\mathbf{X}_{t-119}, \dots, \mathbf{X}_t).$$

Samples are split chronologically to preserve the time-series structure of the data. The in-sample period is further divided into training and validation sets, while the final out-of-sample period is reserved exclusively for model evaluation.

Model Architecture

The forecasting model consists of a deep stacked LSTM architecture with the following configuration:

Component	Specification
Recurrent Layers	5-layer LSTM
Hidden Units	128 per layer
Dropout	0.3 between LSTM layers
Directionality	Unidirectional (causal)
Normalization	Layer normalization
Output	Mean and log-variance of forecast distribution

The LSTM outputs both a conditional mean and conditional variance for future log volatility, allowing the model to represent time-varying forecast uncertainty rather than producing point estimates alone.

Estimation and Training

Model parameters are estimated using the Adam optimizer with a learning rate of 10^{-4} and weight decay of 10^{-5} . Training is conducted for up to 100 epochs, with early stopping applied when validation loss fails to improve for 20 consecutive epochs. To stabilize optimization, gradients are clipped to a maximum norm of 5.0, and the learning rate is reduced when validation loss plateaus.

Rather than minimizing mean squared error, the model is trained by minimizing the negative log-likelihood of a Student-t distribution with fixed degrees of freedom. This loss function accommodates the heavy-tailed nature of volatility innovations and reduces sensitivity to extreme volatility realizations.

3.5 Out-of-Sample Evaluation

All forecasts are aligned with the realized target set and evaluated using:

$$\text{MSE} = \frac{1}{N} \sum_{t=1}^N (y_t - \hat{y}_t)^2$$

$$\text{MAE} = \frac{1}{N} \sum_{t=1}^N |y_t - \hat{y}_t|$$

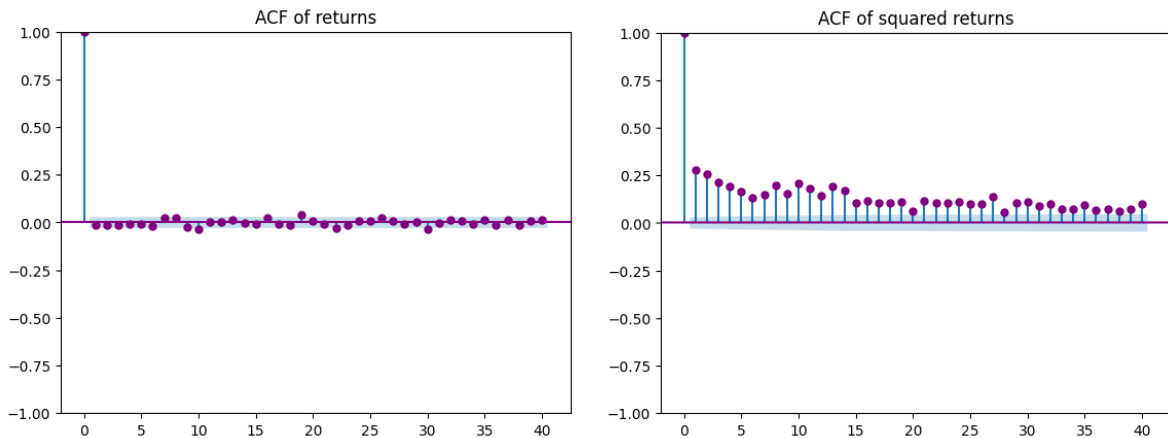
$$\text{QLIKE}_t = \frac{\sigma_t^2}{\hat{\sigma}_t^2} - \log\left(\frac{\sigma_t^2}{\hat{\sigma}_t^2}\right) - 1$$

MSE evaluates the average magnitude of the squared forecast errors, placing greater emphasis on large deviations and providing a strict penalty for periods where the model fails to track volatility spikes. MAE reports the average absolute error and reflects the typical forecast miss in the original scale of the target variable, offering a more interpretable measure of prediction accuracy. QLIKE measures how close a predicted variance is to the realized variance, penalizing both under- and over-estimation in a likelihood-style way (*Patton, 2011*).

4. Results

4.1 Stylized Volatility Dynamics

The autocorrelation patterns in this figure highlight well-established features of equity index returns. The raw log returns display virtually no serial dependence, in line with weak-form market efficiency. In contrast, the squared returns show strong and persistent positive autocorrelation that decays only gradually, reflecting volatility clustering and the long-memory behaviour characteristic of financial market uncertainty. These dynamics justify the use of time-varying volatility models like ARCH and LSTM.



4.2 Out-of-Sample Forecast Comparison

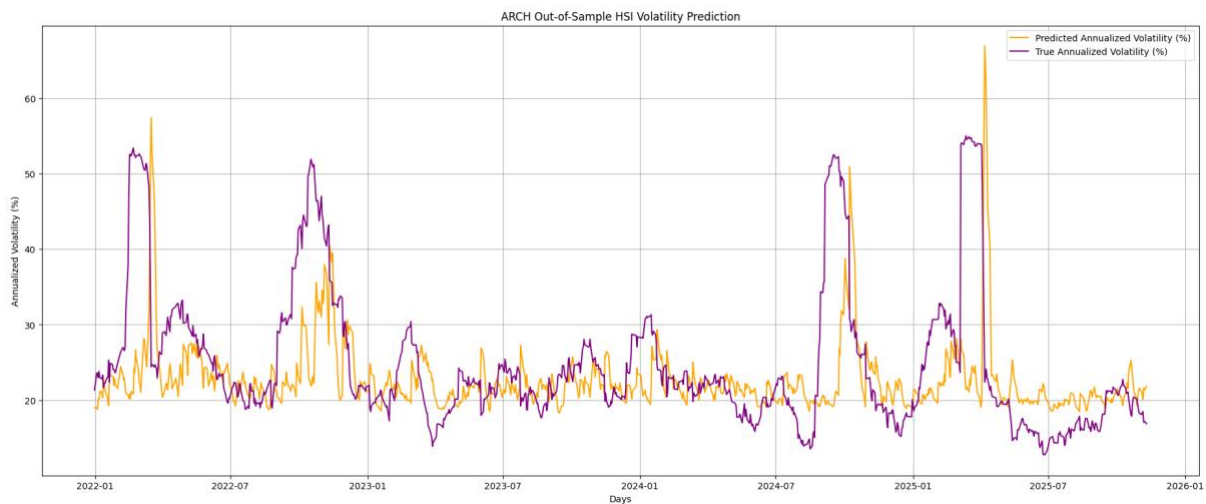
We evaluate forecasts from January 2022 to November 2025.

This table reports quantitative performance metrics:

Model	MAE	MSE	QLIKE
ARCH	0.222	0.097	0.057
LSTM	0.219	0.095	0.055

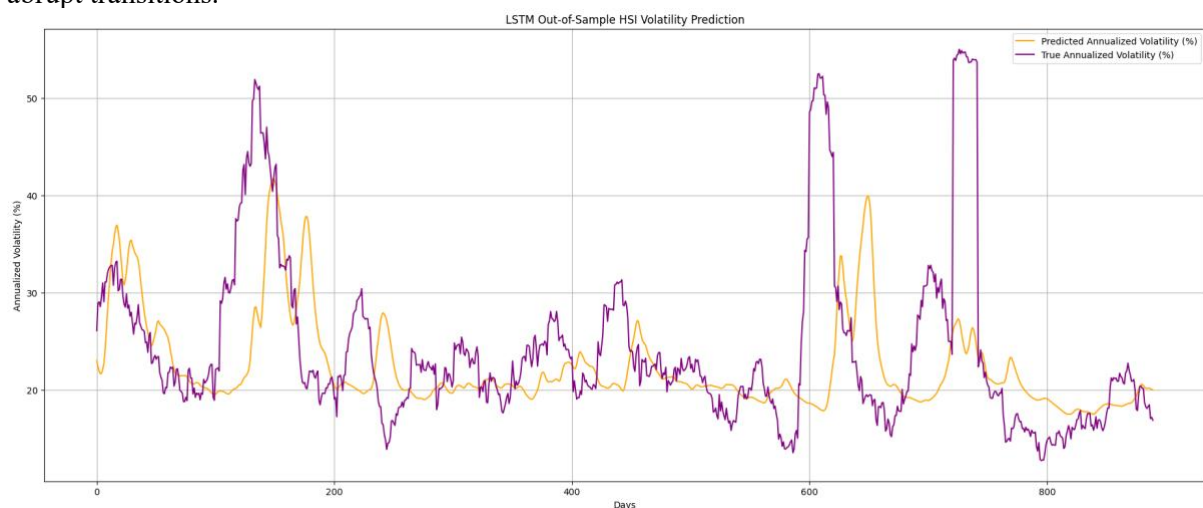
ARCH Forecast Performance

This figure compares the ARCH forecasts with realized annualized volatility over the out-of-sample period. The model captures broad shifts in volatility regimes but responds relatively slowly to abrupt market shocks. Increases in predicted volatility typically occur after large realized spikes and tend to persist during subsequent normalization phases. This delayed adjustment contributes to forecast errors, particularly around sharp volatility transitions. Overall, the standard ARCH structure shows limited ability to adapt to the abrupt volatility swings characteristic of the HSI market.



LSTM Forecast Performance

This figure compares the LSTM forecasts with realized annualized volatility over the out-of-sample period. The model captures the overall level and medium-term movements in volatility reasonably well, particularly during stable market conditions. However, the model exhibits partial smoothing around extreme events. When volatility regimes shift, the magnitude of sharp spikes is often underestimated, and peak volatility is not fully captured. Following large shocks, the LSTM forecasts tend to revert toward the prevailing volatility level with some delay, leading to forecast errors during abrupt transitions.



4.3 Summary of Forecasting Results

The out-of-sample results indicate that the ARCH(7) and LSTM models deliver very similar forecasting performance for 21-day realized volatility of the Hang Seng Index. Across all three evaluation metrics—MAE, MSE, and QLIKE—the differences between the two models are small. While the LSTM achieves marginally lower errors, the magnitude of these improvements is limited and unlikely to represent a material advantage in most practical forecasting applications.

These findings suggest that short-horizon volatility in the Hong Kong equity market is largely driven by recent realized variance, which both models exploit effectively. The ARCH(7) specification captures this persistence directly through lagged squared returns, while the LSTM learns a comparable structure through nonlinear autoregressive dynamics. As a result, both approaches converge to similar predictive accuracy out of sample.

Overall, the results do not support the conclusion that the LSTM meaningfully outperforms the ARCH model in this setting.

5. Economic Interpretation: Volatility and Option Strategies

Volatility forecasting is economically relevant because implied volatility (IV) in option prices reflects the market's expectation of future variance under the risk-neutral measure. Differences between forecasted realized volatility and implied volatility may, in principle, signal potential mispricing. However, the economic value of such signals depends critically on the magnitude and consistency of forecast improvements.

5.1 Forecast Accuracy and Option Pricing

In standard option pricing frameworks, including Black–Scholes and more general stochastic volatility models, option values are increasing functions of volatility. As a result, errors in volatility forecasts translate into option pricing and hedging errors. Forecast evaluation metrics such as MAE, MSE, and QLIKE therefore provide indirect evidence on how well a model aligns with the variance dynamics relevant for option payoffs.

Importantly, small improvements in statistical loss measures do not necessarily translate into economically meaningful gains once transaction costs, bid–ask spreads, and model risk are taken into account. The results therefore caution against interpreting marginal forecast improvements as evidence of substantial pricing or trading advantages.

5.2 From Volatility Forecasts to Trading Signals

In principle, volatility forecasts can be mapped into trading signals by comparing predicted future realized volatility with contemporaneous implied volatility for options of similar maturity. A forecast exceeding implied volatility may suggest a long volatility position, while a lower forecast may suggest a short volatility position.

Such strategies can be implemented using delta-hedged option positions, variance swaps, or volatility index products such as the VHSI. However, given the close similarity in forecasting performance between the ARCH and LSTM models, the implied trading signals generated by the two approaches are likely to be broadly comparable in practice. Any differences in timing or positioning would be incremental rather than structural.

Consequently, the results do not provide strong evidence that the LSTM framework would consistently outperform a traditional ARCH model in volatility trading applications.

6. Conclusion

This paper compared an ARCH(7) volatility model and an LSTM neural network for forecasting 21-day realized volatility of the Hang Seng Index over the period 2022–2025. Using identical information sets and a strictly out-of-sample evaluation, the analysis finds that the two models deliver very similar forecasting performance.

While the LSTM achieves slightly lower MAE, MSE, and QLIKE losses, the differences relative to the ARCH model are small. These results indicate that, for monthly-horizon volatility forecasting at the aggregate index level, a carefully specified linear ARCH model remains competitive with a substantially more complex neural network. The additional flexibility of the LSTM does not translate into a clear or economically meaningful advantage in this context.

Overall, the findings suggest that model complexity alone is not sufficient to guarantee superior volatility forecasts. Future research may examine whether larger gains from machine learning arise when incorporating richer information sets, higher-frequency data, or cross-sectional forecasting problems. In the present setting, however, there is little evidence to favor one approach decisively over the other.

References

- Andersen, T. G., & Bollerslev, T. (1998). Answering the skeptics: Yes, standard volatility models do provide accurate forecasts. *International Economic Review*, 39(4), 885–905.
- Barndorff-Nielsen, O. E., & Shephard, N. (2002). Econometric analysis of realized volatility and its use in estimating stochastic volatility models. *Journal of the Royal Statistical Society: Series B*, 64(2), 253–280.
- Engle, R. F. (1982). Autoregressive conditional heteroskedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4), 987–1007.
- Fischer, T., & Krauss, C. (2018). Deep learning with long short-term memory networks for financial market predictions. *European Journal of Operational Research*, 270(2), 654–669.
- Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8), 1735–1780.
- Patton, A. J., & Sheppard, K. (2009). Evaluating volatility forecasts. In T. G. Andersen, R. A. Davis, J.-P. Kreiß, & T. Mikosch (Eds.), *Handbook of Financial Time Series* (pp. 801–838). Springer.